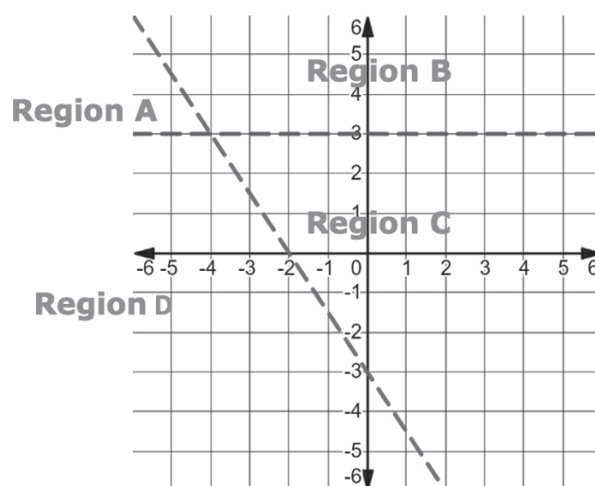


Algebra 1
Unit 6
Lesson 9 Practice

Name _____
Date _____
Period _____

Consider the system of inequalities: $\begin{cases} y < 3 \\ 6x + 4y < -12 \end{cases}$

The boundary lines of both inequalities in the system are shown on the coordinate plane.



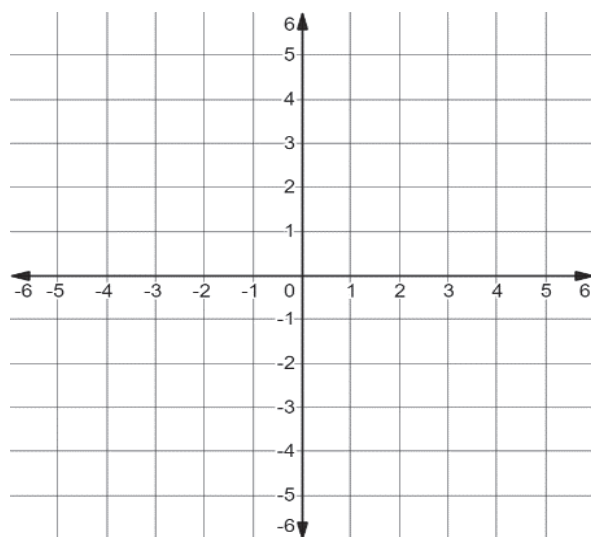
1. The boundary lines separate the plane into four regions, labeled $A - D$. Which region includes the solution set of the system of inequalities?
2. Shade the region that includes the solution set of the system of inequalities.
3. Select a test point in the shaded region of the solution set.
4. Substitute the x -values and the y -values of the test point to show that the ordered pair satisfies both inequalities.

$y < 3$	$6x + 4y < -12$

5. Your friend says that the point $(-4, -3)$ is a solution to the system of inequalities. Explain whether or not you agree.

Consider the system of inequalities: $\begin{cases} y + 2 \geq 0.5(x + 5) \\ y - 3 \leq -3(x + 2) \end{cases}$

6. Graph the boundary lines of the system.



Use substitution to determine whether the given ordered pairs are solutions to the system.

	$y + 2 \geq 0.5(x + 5)$	$y - 3 < -3(x + 2)$	Solution to the system?
7. $(-3, 4)$			☐ Yes ☐ No
8. $(5, 2)$			☐ Yes ☐ No

9. Complete the graph by shading the region included in the solution set of the system.

10. Graph the solution set of the system of inequalities: $\begin{cases} x \leq 2 \\ y > \frac{2}{3}x - 2 \end{cases}$

